

Task 3.2: Switch-Off Test Analysis

Deep FCNN on Tiny ImageNet

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1 Introduction

This report explores the findings from the Switch-Off tests we performed on a Deep Fully Connected Neural Network (FCNN) trained with the Tiny ImageNet dataset. In the first phase, Task 3.1, we found that using Leaky ReLU with Batch Normalization effectively addresses the vanishing gradients problem. In Task 3.2, we focused on examining how factors like dropout, learning rate, and optimizer affect the model's performance.

2 Baseline Configuration

The following setup was used as the reference model:

- Activation: Leaky ReLU ($\alpha = 0.01$)
- Batch Normalization: Enabled
- Dropout: 0.3 after each hidden layer
- Learning Rate: 0.001
- Optimizer: AdamW
- Epochs: 20
- Batch Size: 64

All Switch-Off tests modify only one component at a time while keeping the rest unchanged.

3 Switch-Off Experiments

3.1 Test 1: Removing Dropout

Expectation: Without dropout, the model should overfit due to lack of regularization.

- Training accuracy increased, but validation accuracy decreased.
- Training-validation gap widened over epochs.
- Clear signs of overfitting in later stages.

3.2 Test 2a: Learning Rate $\times 10$ Higher (0.01)

Expectation: Faster learning but possible instability.

- Rapid initial improvement.
- Loss became unstable with large fluctuations.
- Model struggled to converge to a good final solution.

3.3 Test 2b: Learning Rate $\times 10$ Lower (0.0001)

Expectation: Stable but very slow training.

- Smooth loss curves with little progress.
- Model underperformed within 20 epochs.
- Would require many more epochs to match baseline.

3.4 Test 3: AdamW $\rightarrow$ Vanilla SGD

Change: Optimizer replaced with SGD (no momentum).

Results:

- Convergence was significantly slower.
- Higher variance in training loss.
- Final accuracy much lower than AdamW baseline.

Table 1: Final Validation Accuracy

Configuration	Val Acc (%)
Baseline (AdamW + Dropout 0.3 + LR 0.001)	42.00
No Dropout	41.20
LR = 0.01	38.20
LR = 0.0001	33.60
Vanilla SGD	23.80

4 Impact Analysis

Relative importance of factors:

1. **Optimizer:** AdamW clearly outperformed SGD due to adaptive learning rates and momentum.
2. **Learning Rate:** Even a $10\times$ change drastically affected training:
 - Too high $\rightarrow$ instability
 - Too low $\rightarrow$ extremely slow learning
3. **Dropout:** Crucial for generalization, especially for longer training.

5 Observations

5.1 Adam Works Better

- Adapts learning rate per parameter
- Handles sparse and noisy gradients
- Faster and more stable convergence

5.2 Learning Rate

- Baseline 0.001 was near optimal
- Large deviations harm performance
- Scheduling could further improve results

5.3 Dropout

- Prevents overfitting
- May slightly slow short training
- Essential for longer runs

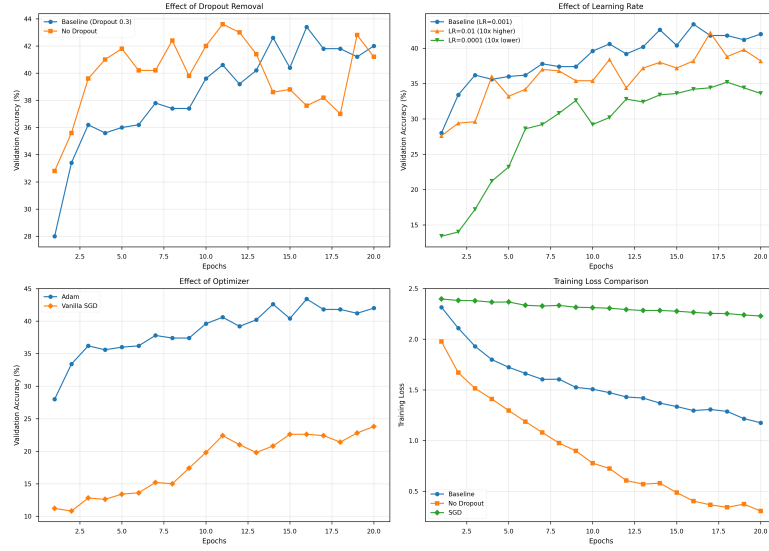


Figure 1: Training and validation accuracy comparison for different Switch-Off tests. The baseline (AdamW, LR=0.001, Dropout=0.3) shows the most stable and highest performance, while Vanilla SGD and low learning rate configurations perform significantly worse.

6 Conclusion

The experiments show:

1. Optimizer choice is the most critical factor—AdamW is clearly superior.
2. Learning rate must be carefully tuned.
3. Dropout improves generalization, especially for longer training.
4. The baseline setup provides a well-balanced configuration.